8 The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} a & -6a & 2a+2 \\ 0 & 1-a & 0 \\ 0 & 2-a & -1 \end{pmatrix}$$

where a is a constant with $a \neq 0$ and $a \neq 1$.

(a) Show that the equation $\mathbf{A}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ has a unique solution and interpret this situation geometrically. [3]

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(b) Show that the eigenvalues of $\mathbf{A}$ are a, $1-a$ and -1. [2]

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(c) Find a matrix **P** and a diagonal matrix **D** such that $\mathbf{A}^4 = \mathbf{PDP}^{-1}$. [6]

(d) Use the characteristic equation of **A** to find $\mathbf{A}^4$ in terms of **A** and a. [3]